SOKOINE UNIVERSITY OF AGRICULTURE
School of Engineering and Technology

Course: ABE 721 - Agricultural Systems Modelling and Simulation
Instructor: Kadeghe Fue, PhD, P.Eng(T)

ASSIGNMENT 3
Module 3: Qualitative and Quantitative Simulation Paradigms

Instructions:

- This assignment tests your comprehension of how mathematical models are actually executed computationally, encompassing numerical integration and parameter estimation.
- Carefully read the ”Significance” statement for each question to understand the practical engineering application of the theory.
- Show all logical steps in your mathematical derivations.
- **Programming Requirement:** For questions that require computational implementation, you must submit your complete, well-commented source code (Python, MATLAB, or R) along with the exact results (console transcripts and any generated plots).

Tier 1: Simple (Conceptual Foundations)

Question 1.1: Time Advancement Paradigms

Significance of Question: *Choosing the wrong time-advancement mechanism results in either massive computational waste (simulating nothing) or critical logical errors.*

Prompt: Distinguish between **Continuous Simulation** and **Discrete-Event Simulation (DES)**. If you were tasked with modelling the exact daily biomass accumulation of a maize crop over a 120-day season, and separately tasked with modelling the queue of tractors waiting to unload grain at a silo, which paradigm would you assign to each problem and why?

Question 1.2: Fuzzy Logic and Qualitative Modelling

Significance of Question: *Capturing human-like decision-making and qualitative variables allows engineers to automate systems where strict deterministic equations fail.*

Prompt: In Fuzzy Logic, how does a **Membership Function** differ fundamentally from strict Boolean logic (1 or 0)? Provide a brief example of how a fuzzy logic rule might be used to control an automated irrigation valve based on a soil moisture sensor reading of "18% volumetric water content".

Tier 2: Intermediate (Application and Logic)

Question 2.1: Numerical Integration Truncation Errors

Significance of Question: *Understanding truncation error prevents numerical drift and explosive instability, ensuring that long-term continuous simulations reflect reality rather than algorithmic noise.*

Prompt: Both the Forward Euler method and the Fourth-Order Runge-Kutta (RK4) method are derived from the Taylor series expansion.

a) What is the Global Truncation Error order for the Euler method, and what is it for the RK4 method?

b) Explain, in your own words, the geometric/mathematical reason why RK4 achieves a much higher accuracy per time-step (h) than the basic Forward Euler method. What does RK4 do differently within the interval t_n to t_{n+1}?

Question 2.2: FDM and CFL Stability

Significance of Question: *In distributed spatial models (PDEs), failing to respect stability criteria will cause the simulation to mathematically explode to infinity, destroying the software's utility.*

Prompt: When solving the 1D Heat/Diffusion equation using the Explicit Finite Difference Method (FDM), engineers are strictly bound by the Courant-Friedrichs-Lewy (CFL) condition.

a) State the CFL condition inequality. If an engineer decides to double the spatial resolution of their grid (halving Δx), what must they geometrically do to the time step (Δt) to maintain stability?

b) Why do modern distributed agricultural models prefer the **Crank-Nicolson method** over the explicit FDM scheme, despite Crank-Nicolson requiring complex matrix inversion?

Tier 3: Hard (Synthesis and Formulation)

Question 3.1: Algorithmic Execution and Programming of Heun's Method

Significance of Question: *Executing the predictor-corrector logic manually bridges the gap between theoretical calculus and actual computer programming algorithms.*

Prompt: Consider a simple exponential decay ODE representing the degradation of a pesticide in soil:

$$\frac{dC}{dt} = -kC$$

Where C is concentration and k is the decay constant.

a) Given an initial condition $C(t_0) = C_0$ and a time step h, write out the explicit mathematical formulas for the **Predictor Step** ($\tilde{C}_{n+1}$) and the **Corrector Step** (C_{n+1}) using Heun's Method (Modified Euler).

b) **Programming Task:** Write a script in Python, MATLAB, or R to numerically solve this ODE using your derived Heun's Method formulas. Let $C_0 = 100$ mg/L, $k = 0.15$ day^{-1}, and simulate the degradation for $t = 0$ to 20 days using a time step of $h = 1$ day. Your submission must include your fully commented code, the console output showing the final concentration at $t = 20$, and a plot of the degradation curve $C(t)$.

Question 3.2: Metaheuristic Parameter Estimation

Significance of Question: *Calibrating highly non-linear agricultural models requires optimization algorithms that do not get trapped in local minima on the error surface.*

Prompt: You are attempting to calibrate a crop model using the Levenberg-Marquardt Algorithm (LMA), but the algorithm continually gets trapped in a "local minimum," yielding a high Sum of Squared Errors (SSE).

a) You decide to switch to a **Genetic Algorithm (GA)**. Describe the specific mathematical/algorithmic mechanism within GA that prevents the population from stagnating and forces it to explore entirely new areas of the parameter map.

b) Define the Objective/Cost Function $J(\mathbf{p})$ that the GA is actively trying to minimize during its "Fitness Evaluation" step.

Please submit your work to my email kadeghe.fue@sua.ac.tz by one week after completion of Module 3. Do not send this work to kadefue@sua.ac.tz.